

SUBJECT	Lower Secondary Science Year 8
TOPIC	Adaptation and Inheritance
ASSIGNMENT	Seven
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE:

- You should complete this assignment by printing out this sheet and then handwriting in black ink, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 40 and 50 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.

LS Science Assignment Seven

VARIATION

1. The drawings show identical twins, Sara and Helen, and their parents.



father



mother



Sara



Helen

a.

- i. Sara and Helen have blue eyes like their mother. Describe how genetic information is passed on from a parent to a child.

(2 marks)

- ii. Sara and Helen have brown hair like their father and blue eyes like their mother. Why do children have characteristics of both parents?

(1 mark)

- b. Sara and Helen are identical twins. Why do they have some identical characteristics?

(1 mark)

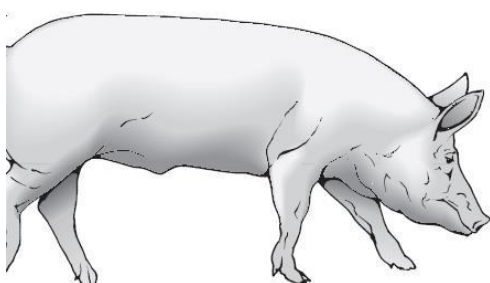
- c. Sara now spends a lot of her time working outdoors in a hot country. Helen now works in an office in England. The table shows information about three human characteristics.

characteristic	Is it identical for Sara and Helen?
eye colour	yes
skin colour	no
weight	no

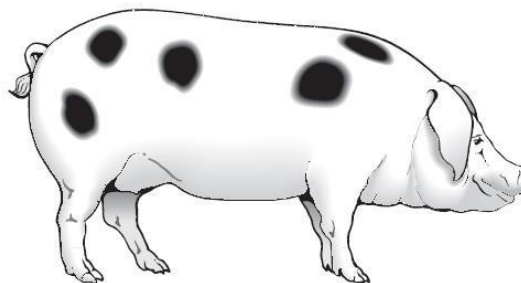
Explain why their eye colour is identical but their weight and skin colour are **not** identical.

(2 marks)
(Total 6 marks)

2. The drawings below show pigs from two different breeds.



Tamworth



Gloucester Old Spot

- a.

- i. From the drawings above, give **two** ways in which the pigs are different.

1. _____

2. _____

(2 marks)

ii. What are these differences called? Tick the correct box.

adaptations

☐

classification

☐

fertilisation

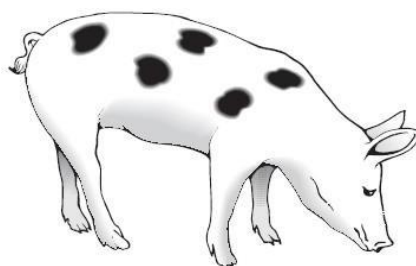
☐

variations

☐

(1 mark)

b. The drawing below shows a piglet bred from a Tamworth and a Gloucester Old Spot.



Give **one** way you can tell that one of its parents is a Tamworth.

(1 mark)

c.

i. When pigs reproduce, which **two** types of cell pass on information from the pigs to their piglets? Tick the **two** correct boxes.

blood
cell

☐

nerve
cell

☐

cheek
cell

☐

egg
cell

☐

muscle
cell

☐

sperm
cell

☐

(2 marks)

- ii. When pigs reproduce, two cells join together. What is this process called? Tick the correct box.

adaptation

☐

classification

☐

fertilisation

☐

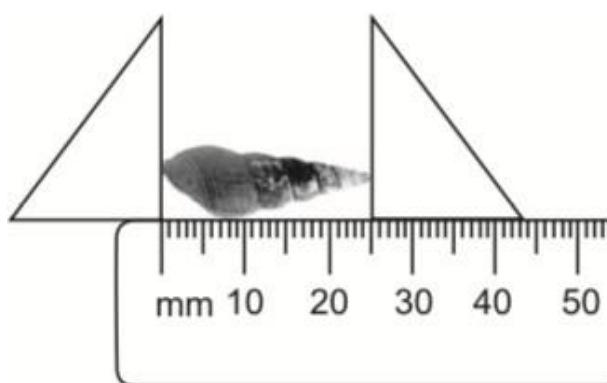
variation

☐

(1 mark)

(Total 7 marks)

3. Jay collected pond snails from the school pond. He measured the lengths of all their shells.



- a. What is the length of the shell above?

_____mm

(1 mark)

- b. Jay made a tally chart of the lengths of **all** the shells he found.

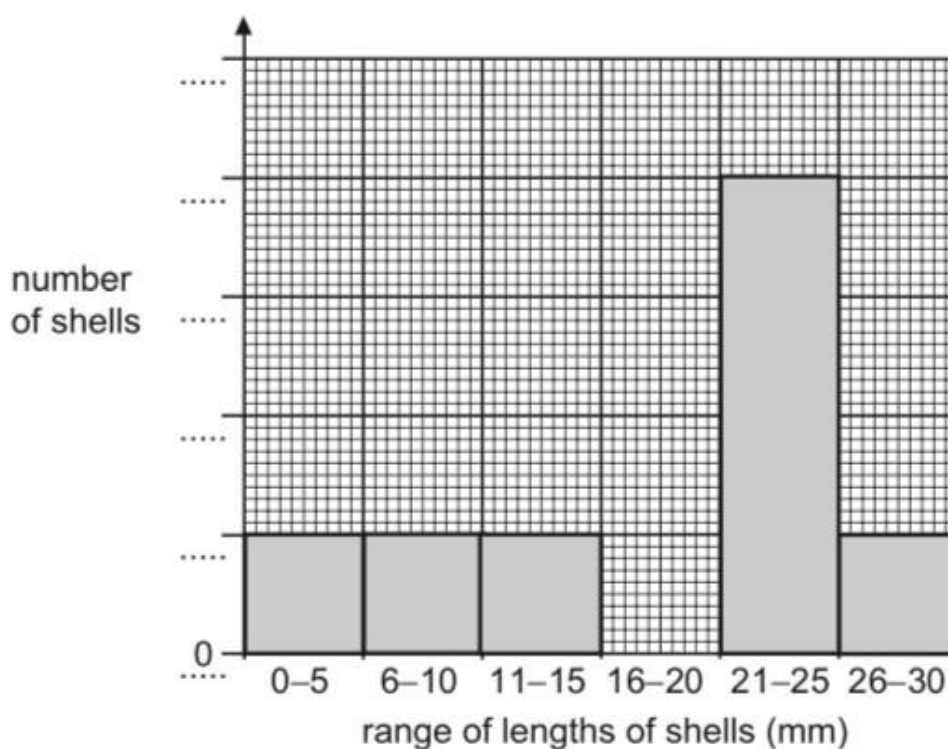
range of lengths of shells (mm)	0-5	6-10	11-15	16-20	21-25	26-30
number of shells	I	I	I	III	IIII	I

What was the most common **range** of lengths of shells Jay collected?

_____mm

(1 mark)

c. Jay recorded his results in a bar chart.



i. Add the missing numbers to the side of the bar chart labelled 'number of shells'.

(1 mark)

ii. **On the chart above**, draw the bar for the number of shells measuring 16-20mm.

(1 mark)

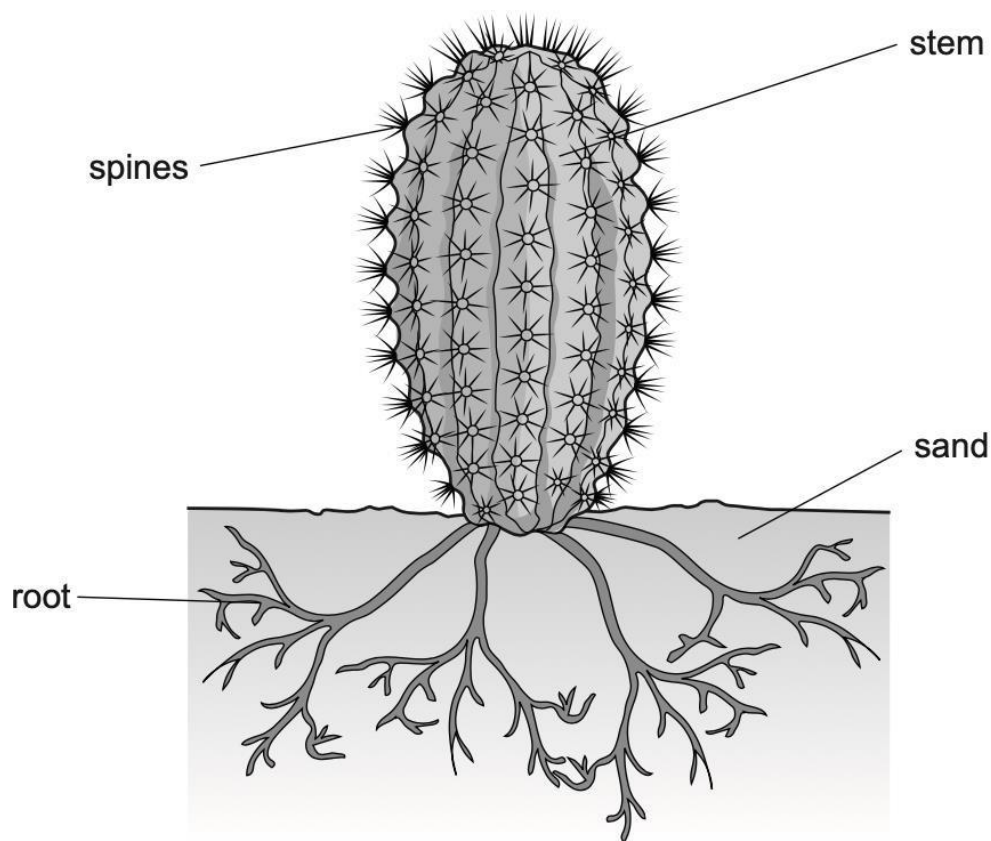
d. Look at Jay's results and decide if each conclusion below is **true** or **false** or if you **cannot tell**. Tick the correct box for each conclusion.

conclusions	true	false	cannot tell
The oldest snails have the darkest shells.	☐	☐	☐
He did not find any shells longer than 30 mm.	☐	☐	☐
He found a total of eight snails.	☐	☐	☐
All the snails he found are the same type.	☐	☐	☐

(2 marks)

(Total 6 marks)

4. This cactus lives in a hot and dry desert.



- a. The cactus has long roots that spread out in the sand.
- i. Suggest how the roots help the cactus survive in the hot and dry desert.

(1 mark)

- ii. Explain one other way the cactus is adapted to living in the hot and dry desert.

(1 mark)

- b. The cactus is eaten by kangaroo rats.
The kangaroo rats are eaten by snakes. Desert foxes feed on snakes.
Complete the food chain for these organism. Include arrows for energy movement.

.....

.....

.....

.....

(2 Marks)

(Total 4 marks)

5. Richard wanted to find out the best conditions for growing lettuce plants.



He took 4 trays and planted 8 lettuce plants in each. The results of his investigation are shown below.

variables				
tray	light level	air temperature (°C)	soil moisture	number of plants alive after 7 days
A	medium	25	moist	8
B	medium	25	dry	6
C	medium	45	moist	2
D	medium	45	dry	0

- a. How many days did Richard's investigation last? Use the table to help you.

_____ days

(1 mark)

- b. Look at the table. Which variables did Richard **change** in his investigation? Tick the correct box.

light level and air temperature

☐

soil moisture and type of soil

☐

air temperature and soil moisture

☐

type of soil and light level

☐

(1 mark)

- c. Richard said:



Lettuce plants grow better at a medium light level than at other light levels.

Why is Richard **not** able to make this conclusion from his investigation?

(1 mark)

- d. The table below shows the number of lettuce plants alive at the end of day 1 and day 7 of the investigation. For each day, A, B, C and D, suggest the number of plants that were alive on **day 4**. Write your answers in the table below.

	number of plants alive		
tray	day 1	day 4	day 7
A	8		8
B	8		6
C	8		2
D	4		0

(2 marks)
(Total 5 marks)

6.

- a. Elephants keep cool by losing heat from their ears.



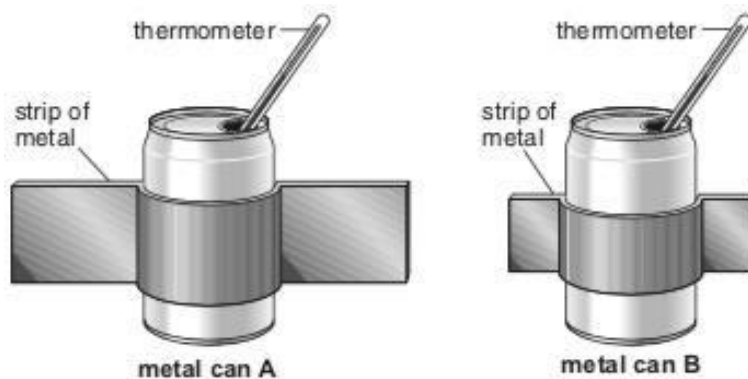
Predict which elephant can lose more heat from its ears.

_____ elephant

Give the reason for your answer.

(1 mark)

- b. Ben filled two identical cans with 250cm^3 of hot water. He wrapped strips of metal around them to model the elephants' ears.



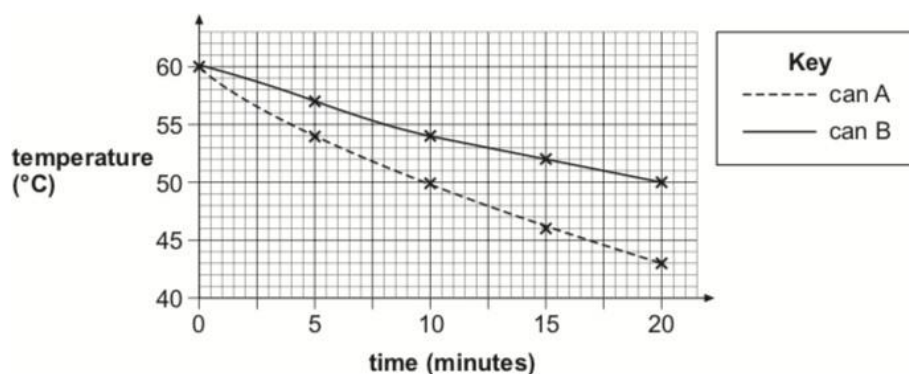
He recorded the temperature of the water in each can every 5 minutes. The table shows his results.

time (minutes)	temperature ($^{\circ}\text{C}$)	
	can A	can B
0	60	60
5	54	57
10	50	54
15	46	52
20	43	50

- i. Ben started with water at the same temperature in both cans. Give **one** other way he made his test fair.

(1 mark)

- ii. He plotted the results for **can A** and **can B** and drew lines of best fit.



Why is it more useful to present these results in a graph rather than a table?

(1 mark)

- iii. The water in **can A** cooled more quickly than the water in **can B**. Does this support your prediction in part (a)? Tick the correct box.

yes ☐ no ☐

Explain your answer.

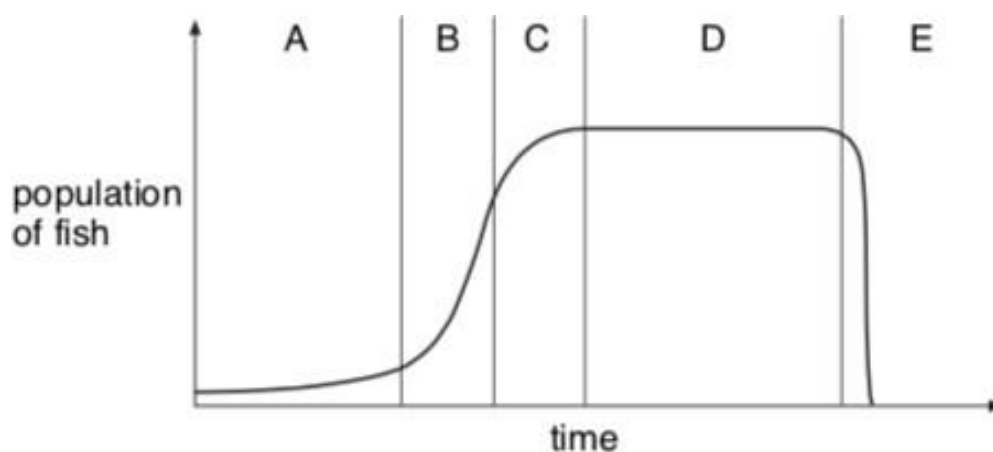
(1 mark)

- c. Ben repeated the investigation. Instead of a thermometer he used a temperature sensor and a data logger. Give **one** advantage of this.

(1 mark)

(Total 5 marks)

7. The graph below shows how a population of fish in a lake changed over a period of time.



- a. In which time interval, A, B, C, D or E, did the population of fish increase most quickly?

How can you tell this from the graph?

(1 mark)

- b. Which part of the graph shows when the fish **began** to compete with each other for food? Give the letter.

How can you tell this from the graph?

(1 mark)

- c. What does part D of the graph show about the birth rate and the death rate of the fish?

How can you tell this from the graph?

(1 mark)

- d. Part E of the graph shows a population crash when all the fish died. Suggest **two** reasons why a population might crash in this way.

1. _____

2. _____

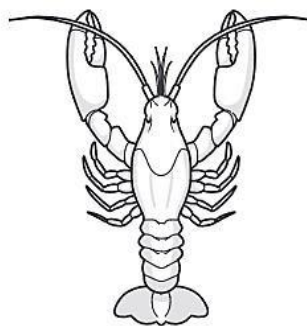
(2 marks)

(Total 5 marks)

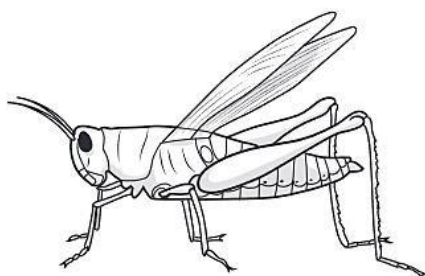
8. Look at the diagrams of the four arthropods **A**, **B**, **C** and **D**.



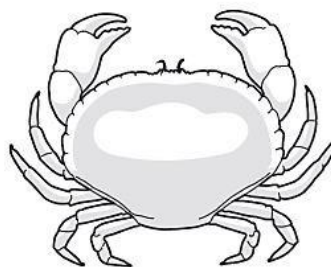
A



B



C



D

NOT TO SCALE

Use the key to identify each arthropod.

1a	has two claws	go to 2
1b	does not have claws	go to 3
2a	has a segmented tail	<i>Orconectes punctimanus</i>
2b	does not have a segmented tail	<i>Cancer pagurus</i>
3a	has wings	<i>Schistocerca gregaria</i>
3b	does not have wings	<i>Enantiulus armatus</i>

Write the letter of the correct arthropod next to its name.

Orconectes punctimanus _____

Cancer pagurus _____

Schistocerca gregaria _____

Enantiulus armatus _____

(4 marks)

TOTAL FOR ASSIGNMENT 42 MARKS